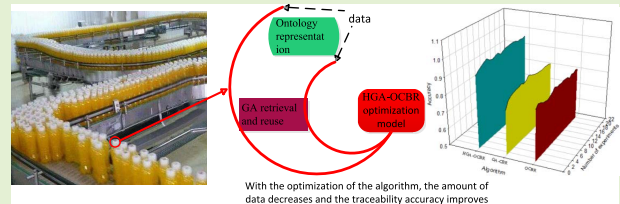


An Optimization Model for Process Traceability in Case-Based Reasoning Based on Ontology and the Genetic Algorithm

Yineng Chen^{ID}, Xinghui Zhu^{ID}, Kui Fang^{ID}, Yanbin Wu, Yuechao Deng^{ID}, Xiao He^{ID}, Zhuoyang Zou^{ID}, and Ting Guo^{ID}

Abstract—A traceability system can quickly and accurately query product supply chain circulation information. It is difficult to obtain processing information and achieve sharing due to the big data, long duration, large amount of equipment and scattered and easily lost processing information. In tracing the application of the case-based reasoning (CBR) to product processing, we can solve the problem of establishing an accurate and complete mathematical model between the processing information and product quality defects by finding a target record in a large amount of historical data. The traceability model of ontology and CBR uses the web ontology language (OWL) for case representation to simplify the CBR framework for big data, classifies process data and standardizes information storage. A joint optimization algorithm based on the genetic algorithm (GA) and CBR is proposed to establish an optimization model of process traceability, which is applied to case retrieval and reuse. This algorithm optimizes the genetic operator by combining the optimal preservation strategy and roulette selection method and uses an exponential-scale transformation method to stretch the fitness function. The experiments show that the optimized traceability model can infer information from garbled codes, wrong codes and missing messages to quickly determine the problematic products, thus effectively improving the traceability accuracy of product processing.

Index Terms—Ontology, GA, traceability optimization, CBR, processing information.



I. INTRODUCTION

THERE WAS a global outbreak of COVID-19 (coronavirus disease 2019) in 2020. The virus has been detected in cold chain fresh products, but it is unknown at which production link the virus was introduced. Moreover, it is impossible to accurately find staff and consumers who have been exposed

to virus-infected products, and it takes time and effort to investigate contacts on a large scale. However, a traceability system can accurately find contacts and determine the links of product infection, which plays an important role in virus protection and product safety. CAC (Codex Alimentarius Commission) and ISO (International Standardization Organization) have defined traceability, which is the ability to trace the history of behaviors and the use or location of goods through a registered identification code [1]. The traceability system can quickly and accurately query all information in the supply chain. When product safety and quality problems occur, the problematic products and their causes can be quickly found through the traceable source code on the product label. This code is specific to an origin, warehouse, distribution route, inspection and employee. As people pay increasing attention to food quality and safety issues, the traceability system has not only become a part of product trade and development but is also inseparable from people's daily lives and the economic development of countries. A large number of studies have shown that the use of traceability systems in processing and sales can achieve improved results.

In December 2019, Jiangnan University and China Food Safety News jointly released the "China Food Safety Development Report 2019" report in Beijing. The report noted

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that the food safety incidents that occurred in 2018 were mainly in production and processing, accounting for 45.16% of the total [2]. It is very important for product quality and safety to trace the source accurately in production and processing. Pizzuti *et al.* [3] described an ontology developed for supporting the management of meat traceability along a whole supply chain. MESCO is able to represent all the knowledge and information related to the meat traceability domain in a single ontology, enabling interoperability among different systems. This method would be better if it could make automatic decisions. Wang *et al.* [4] improved the traceability of wheat flour tracking information and the efficiency of hazard queries and reduced the scope of defective product recall. However, this method is not suitable for traceable data with high complexity and similarity. Chen [5] proposed a traceability method based on intuition and fuzzy case-based reasoning and adopted a sorting method to predict the performance similarity of products. This method can distinguish traceable information with high similarity but cannot quickly classify traceable products. Zhang *et al.* [6] proposed a case-based reasoning method for the traceability of exhaust gas pollution and constructed a general ontology model to represent exhaust gas traceability cases, but the attributes of the cases need to be optimized. Gautam *et al.* [7] implemented a plant pollinator optimization algorithm to reduce the cost of logistics, implementation and liability for fruit traceability. This method can find the grower or pack house that pays the cost of examination, but it cannot determine which grower or pack house performs the examination. Bin *et al.* [8] proposed a solution to the optimization problem of batch mixing for the traceability of fresh-cut vegetable processing, which optimized the order and finished product processing sequence and raw material selection sequence. However, this method does not take the traceability information of the processing environment into account, such as processing equipment. Chen [9] used case-based reasoning to optimize the production process, which has strong practicability, but the case representation method of industrial production process attributes needs to be further studied. Khan *et al.* [10] proposed a model to optimize the data of the food supply chain based on a recurrent neural network. A genetic algorithm was used to optimize the parameters of the model, which had a very good effect. With the development of traceability systems, production and processing traceability has four research phases: origin traceability, quality traceability, resource optimization traceability and precise traceability, as shown in Figure 1. In general, references 3-6 are methods that study origin traceability and quality traceability, references 7 and 8 are methods that study resource optimization in processing, and references 9 and 10 are methods that optimize processing. However, studies on precise traceability optimization for processing are rarely reported.

To improve the traceability accuracy of the product supply chain, a processing traceability model is constructed by CBR for complex product processing links. A joint optimization method based on ontology, the genetic algorithm and case reasoning is proposed to optimize the model. This method is based on ontology for case representation, while the improved genetic algorithm is applied to case retrieval and case reuse.

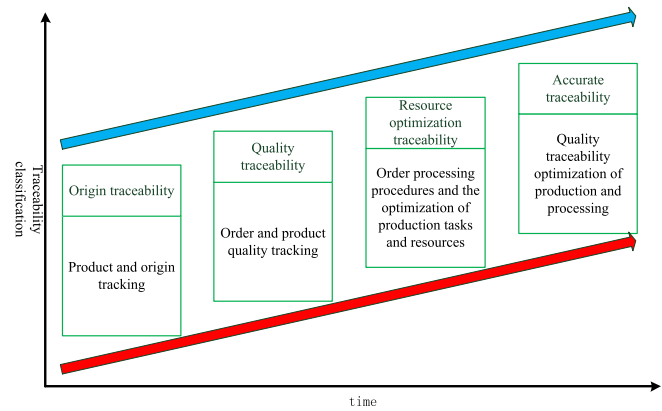


Fig. 1. Research progress in production and processing traceability.

It effectively improves the traceability accuracy of product processing and ensures the integrity of traceability data.

II. CASE REASONING FOR PRODUCT PROCESSING TRACEABILITY

The traceability information of the product processing link includes information on the raw materials, order, equipment, finished products and other relevant information. It is not only difficult to record and collect data but also error-prone in data acquisition, which may cause confusion in product supply chain data and even affect the accuracy of the system traceability information [11]. In addition, the traceability information will change with product processing [12]. Therefore, to trace the processing history of a product, it is necessary to trace all the equipment and its environment in the whole process to trace the complete information of the supply chain.

According to the time, batch, name, number and sequence of warehousing, a new batch (material batch, MaB) will be generated for the raw materials purchased from the processing plant. After receiving the raw materials, the processed products are formed through key processing steps. However, the processing steps are complex and uncontrollable. To solve this problem, the processing information is based on processing equipment in this paper. RFID technology is used to associate the MaB with a processing batch (PiB). The generated RFID tag is printed on the packaging bag or other transfer device for identification, including the equipment number, equipment name, date, time, temperature, and pressure. After processing, the finished product batch (PoB) is formed, which is composed of the packing date, product name, product quality and sequence. The map of the product processing information is shown in Figure 2.

The MaB, PiB (processing equipment information) and PoB are recorded in sequence and closely related to each other. The processing information to be queried can be inferred from the relevant historical data. CBR [13] is used to solve new problems by looking for specific knowledge in the experience or results of similar historical cases. It has the advantages of being a simple, fast and efficient problem solving method. This method can transform knowledge, experience and models that are difficult to regularize in historical records into intuitive understandings [14]. Product processing traceability

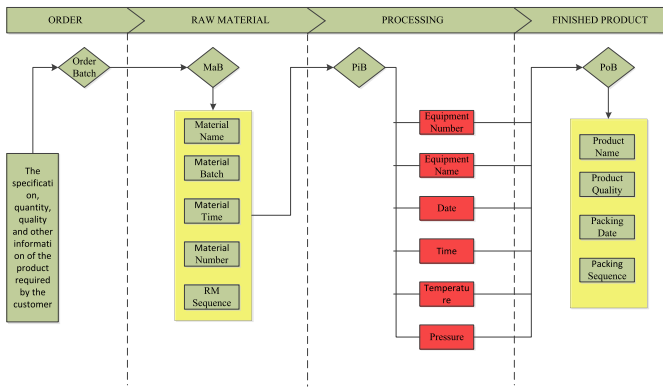


Fig. 2. Map of the product processing information.

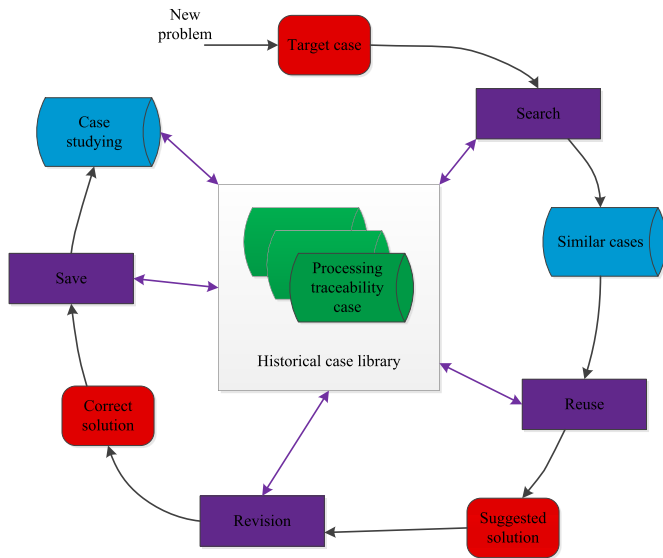


Fig. 3. The process of CBR for product traceability.

determines the product processing information according to the finished product batch number of the problematic product and then finds the cause of the product quality defect. It is difficult to establish an accurate and complete mathematical model between the processing information and product quality when there are a large number of historical processing information defect records. Therefore, it is very desirable to apply CBR to traceability in product processing. The process of CBR for product traceability is shown in Figure 3.

However, due to the large amount of data, strong time continuity and large amount of production equipment, the processing information acquisition and sharing efficiency are low, which brings great difficulty for product traceability [15]. On the other hand, raw material information, warehouse information, and processing information will not be managed in a unified manner. Recording and storage are carried out separately, causing product processing information to be scattered or lost, and product processing traceability cannot be performed normally. Therefore, establishing processing information records and storage standards to provide a basis for management will be conducive to improving the efficiency of information acquisition and sharing to improve the product traceability accuracy from the data source. To standardize

and classify processing information recording and storage, ontology will be applied to case representation and retrieval, which can simplify the big data CBR framework.

III. ONTOLOGY-BASED TRACEABILITY CASE REPRESENTATION MODEL CONSTRUCTION

Ontology can not only be suitable for the definition and representation of product processing traceability cases but can also be integrated into the general knowledge of the processing field. Moreover, ontology has a more mature modeling language, query language and development tools [16]. It is convenient to retrieve and modify the product processing traceability cases represented by ontology.

Case representation mainly adopts eigenvectors or structures, both of which have advantages and disadvantages [17]. The advantage of an eigenvector representation is that it is simple and effective, and the disadvantage is that the utilization rate of processing data is not high enough to improve the retrieval efficiency of cases [18]. The advantage of a structured representation is that it makes full use of the knowledge of the target case domain [19], such as traceability classification knowledge. Since a structured representation uses a semantic network or framework to define the case structure and knowledge, the two knowledge representations are nonformal representation methods, which bring many difficulties for the storage, modification and retrieval of cases. It is not only difficult to share case knowledge and domain knowledge but also complicated to manage and maintain the case base.

As a formal conceptual model, ontology can define both the eigenvector and the structured representation method of the case. In the ontology of knowledge representation logic based on description logic (DL), DL is a decidable logical representation language that formalizes the description of deterministic, shared knowledge within a domain [20]. According to the characteristics of product processing traceability, ontology-based case representation can be defined by the following triple:

$$\text{Case} = \langle D, C, S \rangle$$

where D is the definition set, C is the case set, and S is the solution set.

A. Case Structure Definition

For the case of eigenvector representation, the OWL-DL abstract syntax is adopted, where na represents a custom namespace prefix, $i \in [1, n]$, and n is the dimension of the eigenvector. In the structured case, an interfeature relationship that is used to represent the inclusion relationship is added based on the eigenvector approach. The case structure definition steps for processing information traceability are shown in Figure 4.

According to the definition steps of the case structure, the feature vector of the case of processing information traceability can be simplified as $\langle \text{MaB}, \text{MaN}, \text{PD}, \text{QI}, \text{Qt}, \text{ENu}, \text{ENa}, \text{PT}, \text{PoE} \rangle$, where MaN refers to the material name, PD refers to the processing date, QI refers to the quality, Qt refers to the quantity, ENu refers to the equipment number, ENa refers to

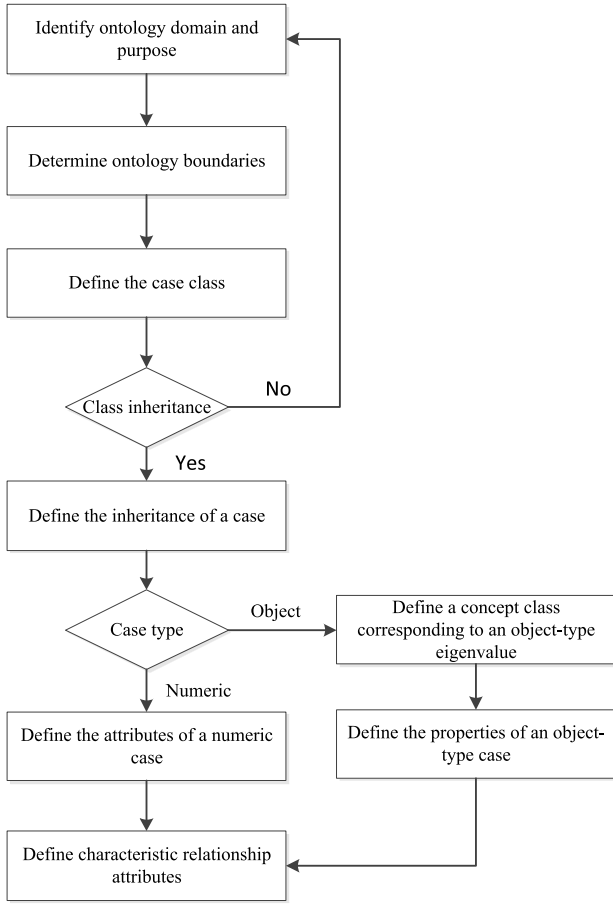


Fig. 4. The case structure definition steps for processing information traceability.

the name of the equipment, PT refers to the processing time, and PoE refers to the processing environment. It should be noted that the processing time refers to the time spent in the corresponding process of the equipment processing products, and the processing condition refers to the temperature, pressure and other factors in the processing process. Therefore, PD, Qt and PT are numerical properties, while MaN, Ql, ENa and PoE are object properties. According to the DL case representation method, it can be expressed as:

Clacc(na: Processing traceability partial owl:Thing);

Clacc(na: MaN partial owl:Thing);

Clacc(na: Ql partial owl:Thing);

Clacc(na: ENa partial owl:Thing);

Clacc(na: PoE partial owl:Thing);

Datatype Property (na: MaB domain (ex: Processing traceability));

Datatype Property (na: PD domain (ex: Processing traceability));

Datatype Property (na: Qt domain (ex: Processing traceability));

Datatype Property (na: ENu domain (ex: Processing traceability));

Datatype Property (na: PT domain (ex: Processing traceability));

Object Property (na: has Materialname domain (na: Processing traceability) range (na: MaN));

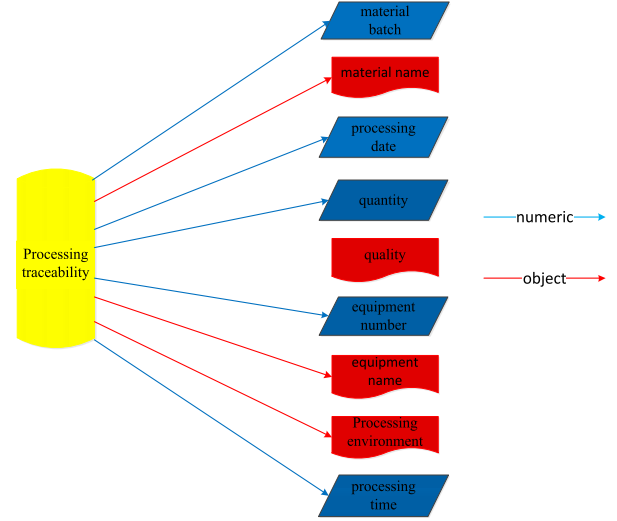


Fig. 5. The case structure definition for processing traceability.

Object Property (na: has Quality domain (na: Processing traceability) range (na: Ql));

Object Property (na: has Devicename domain (na: Processing traceability) range (na: ENa));

Object Property (na: has processing environment domain (na: Processing traceability) range (na: PoE));

The definition domain of the Datatype Property and Object Property is the processing traceability, and the value domain of the Object Property is the corresponding concept class. For example, the value domain of 'has Materialname' is 'raw material product name'. Figure 5 is the case structure definition for processing traceability.

B. Representation of a Case Instance

After defining the case representation and structure, the value of the case instance is assigned to the corresponding feature item. If it is an object-type attribute, this attribute can only be added between two individuals in DL. Therefore, an eigenvalue is an instance of a class, which is assigned by attaching an attribute between the case instance and the eigenvalue instance.

IV. CASE RETRIEVAL OPTIMIZATION MODEL CONSTRUCTION BASED ON A GENETIC ALGORITHM

During product processing, many different products may be produced by the MaB. However, when the processing flow and the raw material ratio are consistent, the same equipment is used to process these different products, and their processing environments are also similar. Therefore, there are too many data points for product processing, and they are very similar. In this case, it is difficult to retrieve the historical case that matches the target case with the traditional case retrieval or ontology similarity calculation method. To improve the accuracy of case retrieval, this paper uses the genetic algorithm to optimize the case retrieval model. The genetic algorithm is a kind of intelligent search algorithm that has all the characteristics of intelligent search [21]. In addition, it is a global random search algorithm [22]. In contrast, using the

TABLE I
BINARY BITS OF THE CASE ATTRIBUTE ENCODING

Encoding	MaB	MaN	PD	Qt	Ql	ENu	ENa	PT	PoB
weight	1000	50	100/12/31	1000	10	60	30	2359	1000
binary bits	10	6	16	10	4	6	6	12	10

genetic algorithm to retrieve historical cases can effectively improve the reasoning speed.

The genetic algorithm replaces the parameter space of the problem with the coding space, takes the fitness function as the evaluation, and takes the coding group as the evolution basis [23]. An iterative process is established to carry out the selection and genetic mechanism of individual bit strings in the population. New excellent individuals are searched randomly in the solution space through chromosome crossing and mutation of the individuals in the population until an optimal or suboptimal solution that satisfies the condition of convergence is found [24]. The principle and execution of the genetic algorithm are shown in Figure 7.

A. Coding Scheme Design

The optimization solution of the genetic algorithm can only be obtained by encoding the parameters that need to be optimized; that is, the weights of the case attributes and cases need to be coded. Due to the high complexity and similarity of product processing traceability cases, the binary coding method is adopted in this paper, which can quickly distinguish each case by simplifying genetic operations such as crossover and mutation.

The population in the genetic algorithm is composed of individuals, each of whom represents a solution to a problem, that is, a historical case. The more historical cases there are in the population, the stronger the search ability of the algorithm, and the more likely it is to obtain the optimal solution, but the calculation time will be longer. The more binary digits there are, the wider the range of attribute weights that can be expressed, but the computational complexity of the algorithm will increase. Therefore, this article adopts the case weight attribute strategy, in which the attribute of each individual includes the attribute of the case. The number of bits in the binary code is designed separately according to the range of each property, for a total of 76 bits. The setting of the number of bits is shown in Table I.

In the table, the date (20201231) occupies too many binary digits, so it can be reduced by dividing it into the year (20), month (12) and day (31). The processing time is the specific time based on the processing date, and the range is 00:00-23:59. To obtain the genes of the genetic algorithm, the value of the case attribute is converted into the corresponding binary value. Examples of populations are shown in Table II.

B. Selection of the Fitness Function

To evaluate the fitness of each individual, it is necessary to make an inference decision regarding the accuracy of the CBR

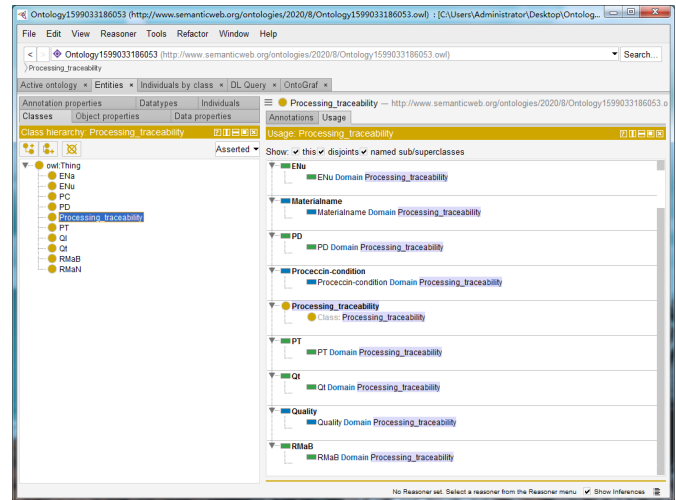


Fig. 6. Case example of processing traceability.

system determined by the individual based on the training sample. The specific steps are as follows:

Step 1: For the current individual, randomly select K cases from the original case library as the training set C. The attribute weight of each case, that is, the problem description in set C, is used as the input of case reasoning.

Step 2: Based on the case reasoning system for the current individual, perform reasoning operations on set C to obtain the solution of the training case.

Step 3: Compare the solution obtained by the inference with the real solution of the training case, and obtain the decision accuracy of the individual for the k training cases. The maximum value of the sum of the decision accuracy is selected as the fitness function.

To improve the computational efficiency and precision of the genetic algorithm, real number coding is adopted for each attribute. Express each group of encoded weights as an individual case in the population, which can be expressed as

$$C_i = (c_{i1}, c_{i2}, c_{i3}, c_{i4}, c_{i5}, c_{i6}, c_{i7}, c_{i8})$$

Through CBR reasoning, the decision accuracy of the genetic algorithm for processing traceability cases is shown in Table III.

In the table, c_{ij} represents the weight of individual c_i for attribute j . The higher its value, the more important the attribute is. D is the decision accuracy of the current individual in the training case, which can be expressed as

$$D = \begin{cases} 1, & \text{if } c_{ij} = C_{ij} \\ 0, & \end{cases}$$

TABLE II
EXAMPLE OF A PRODUCT PROCESSING TRACEABILITY POPULATION

A	Attribute weight					
	w_1	w_2	w_3	...	w_8	w_9
1	0000000000	000000	0000000000000000	...	000000000000	0000000000
2	0000000001	000001	0000000000000001	...	000000000001	0000000001
3	0000000010	000010	0000000000000010	...	000000000010	0000000010
...				...		
n	1111111111	111111	1111111111111111	...	111111111111	1111111111

TABLE III
CALCULATION OF THE DECISION ACCURACY OF THE GENETIC ALGORITHM

Case	Decision value of the training case				True value of the training case				Accuracy
case ₁	c_{11}	c_{12}		c_{18}	C_{11}	C_{12}		C_{18}	D_1
case ₂	c_{21}	c_{22}		c_{28}	C_{21}	C_{22}		C_{28}	D_2
.....									
case _k	c_{k1}	c_{k2}		c_{k8}	C_{k1}	C_{k2}		C_{k8}	D_k
$f(C)$									

$f(C)$ is the total decision-making accuracy of the current individual on training set C , which is selected as the fitness function for judging the individual, which can be expressed as

$$f(C) = \sum_{i=1}^k D_i$$

The larger the F value is, the higher the decision-making accuracy of the case-based reasoning system determined by the current individual, and the more likely the individual is to be inherited by the next generation. Take the maximum value of fitness as the objective function of the genetic algorithm.

C. Genetic Algorithm Generates a New Population

The selection operation in the genetic algorithm selects good individuals from the parent population and inherits them in the next-generation population. The selection operation is used to determine the recombination or crossover as well as how many offspring will be produced by the selected individuals [25]. The selection operation is not only used to determine the individuals in the recombination or crossover but also calculates how many offspring individuals to produce. Although genetic algorithms can produce an increasing number of excellent individuals, they are likely to destroy the most adaptive individuals in the current population due to the randomness of the genetic operators. Of course, the efficiency and convergence of the genetic algorithm are affected. This paper adopts the best retention selection method to retain the most adaptable individuals in the next-generation group [26].

In this method, the individual with the highest fitness does not participate in the operations of crossover and mutation, but instead it is used to replace the lowest-fitness individual after crossover and mutation in the current generation population. Combining the optimal preservation strategy with the conventional roulette selection method will make the optimization process of the genetic algorithm efficient and accurate. The realization process is shown in Figure 8.

For an individual C_i ($i = 1, 2, 3, \dots, N$), the individual's selection probability p_i is

$$p_i = \frac{f(C_i)}{\sum_{a=1}^N f(C_a)}$$

where $\sum_{a=1}^N f(C_a)$ is the sum of probabilities in the population.

To avoid precocity and low diversity in the multigeneration evolution algorithm, the fitness function is stretched by the exponential scale transformation method. In the initial stage of evolution, it narrows the difference in fitness between individuals to maintain diversity in the population. In the later stage of evolution, it enlarges the difference in fitness to make the performance of the good individuals more prominent. The selection probability P_i of individual C_i is improved to

$$p'_i = \frac{e^{\frac{0.5gf(C_i)}{G}}}{\sum_{a=1}^N e^{\frac{0.5gf(C_a)}{G}}}$$

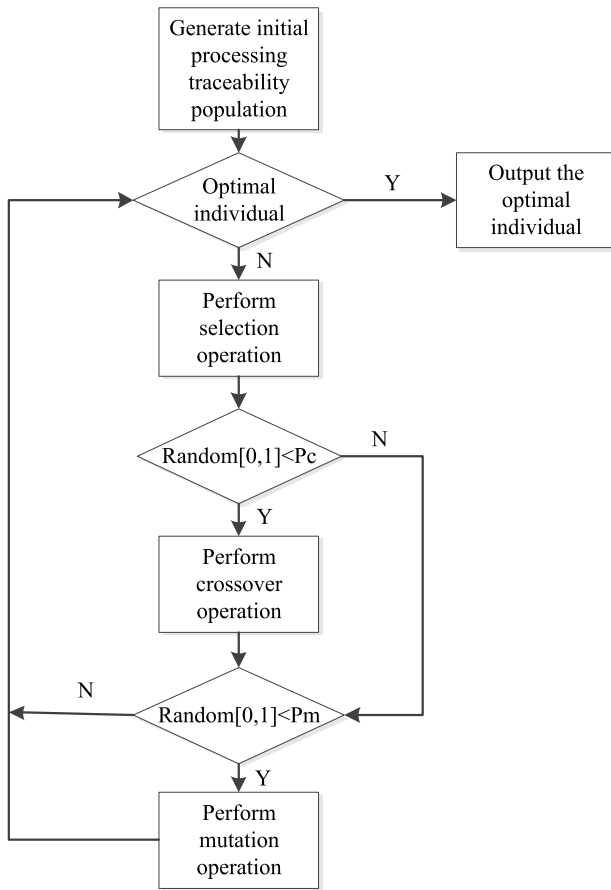


Fig. 7. The principle and execution of the genetic algorithm.

where g is the current evolutionary generation and G is the total number of evolutionary generations of the population.

To select a good individual and ensure that it is outstanding in each generation of evolution, we not only consider its contemporary selection probability but also consider the evolutionary historical selection probability. The worst individual $\text{Min}(\bar{p}_i(h+1))$ is replaced in the next generation with the best contemporary $\text{Max}(\bar{p}_i(h))$. The historical selection probability $P_i(h)$ of an individual C_i is defined as

$$\bar{p}_i(h) = \frac{\sum_{h=1}^g p_{ih}}{\sum_{h=1}^g h}$$

where p_{ih} is the selection probability of C_i in generation h .

V. EXPERIMENTAL RESULTS AND DISCUSSION

A. Experimental Design

We collected 1000 sets of processing data from a beverage processing factory. Based on these data, the case structure set, case set and solution set are established on Protégé. Then, we used Jena to save the OWL file of the beverage ontology to the MySQL database to establish a traceability case and designed a genetic algorithm in Jena. Additionally, the genetic algorithm was designed in Jena to complete the reasoning

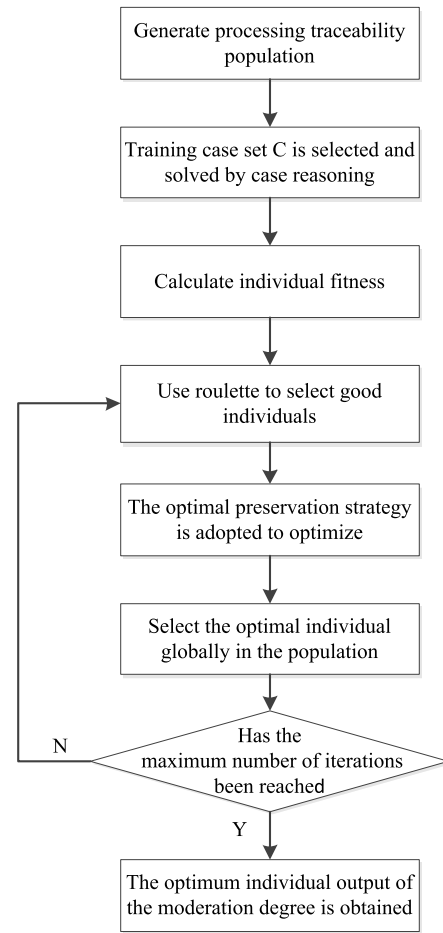


Fig. 8. The process of optimizing the genetic algorithm based on the optimal preservation strategy and the roulette selection method.

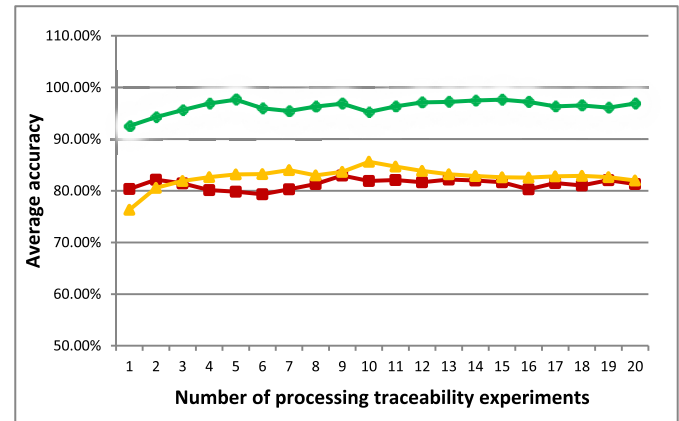


Fig. 9. Average accuracy of traceability of the 3 methods.

of the traceability case. Table IV shows an example of some traceability cases of beverage processing.

To verify the performance of the method, the following three groups of comparative experiments were used to trace the beverage processing information:

- 1) The ontology model was used to represent the case, and the K-nearest neighbors (KNN) algorithm was used to perform case retrieval, which was called OCBR.
- 2) A genetic algorithm was used to optimize the weight of the CBR attributes and to reduce the case base, which was called GA-CBR.

TABLE IV
SOME EXAMPLES OF TRACEABILITY CASES IN BEVERAGE PROCESSING

Case	MaB	MaN	PD	Qt	Ql	ENu	ENa	PT	PoB
1	100	sydney	0323	50 kg	1	5	centrifuge	9:06	15 MPa
2	101	sydney	0323	20 kg	1	5	centrifuge	11:12	15 MPa
3	102	sydney	0324	10 kg	1	8	screw juicer	9:16	15 MPa
4	103	sydney	0324	50 kg	1	8	screw juicer	10:04	15 MPa
5	200	papaya	0323	50 kg	1	24	homogenizer	8:34	30 MPa
6	201	papaya	0324	20 kg	1	24	homogenizer	8:26	30 MPa
7	202	papaya	0323	10 kg	1	27	sterilizer	10:46	30 MPa

3) The ontological model was used to represent the case. Additionally, the optimal historical selection improved genetic algorithm was used to optimize the weight of the CBR attributes and to reduce the case base, which was called HGA-OCBR.

The experimental parameter settings were as follows: There were 1000 cases in the traceability case library for beverage processing, with 1 bit of binary code for each case. The case had eight feature attributes, with a 10-bit binary code representing one feature attribute weight. Thus, the individual chromosome length was 1080 digits. For the current individual, 50 cases were selected from the case base species as the training set. The five parameters of the genetic algorithm were as follows: the initial population size N was 100, the maximum iteration number G was 300, the crossover probability P_c was 0.7, the variation probability P_d was 0.05 and the K value was 3.

B. Experimental Results

To verify the accuracy and traceability speed of the optimization model, OCBR, GA-CBR and HGA-OCBR were used to conduct comparative experiments on 200 sets of selected data, and the average accuracy rate, accuracy error, accuracy stability, average traceability time, and traceability speed stability were recorded.

Figure 9 is a comparison chart of the average accuracy of HGA-OCBR, OCBR and GA-CBR. The average accuracy of HGA-OCBR is higher than OCBR and GA-CBR. OCBR has the lowest average accuracy of traceability. As the number of traceability experiments increases, the average accuracy of HGA-OCBR stabilizes at approximately 96%-98%.

Figure 10 shows the error rates of the three methods. The error of HGA-OCBR is much lower than that of OCBR and GA-CBR. OCBR and GA-CBR have similar errors, but the error rate of GA-CBR is slightly lower than that of OCBR. As the number of traceability experiments increases, the error of HGA-OCBR decreases by approximately 2%-4%, which also shows that the average accuracy of HGA-OCBR is the highest.

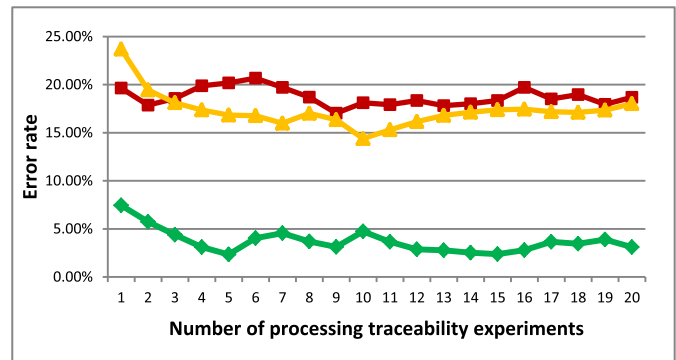


Fig. 10. Error rate of the 3 methods.

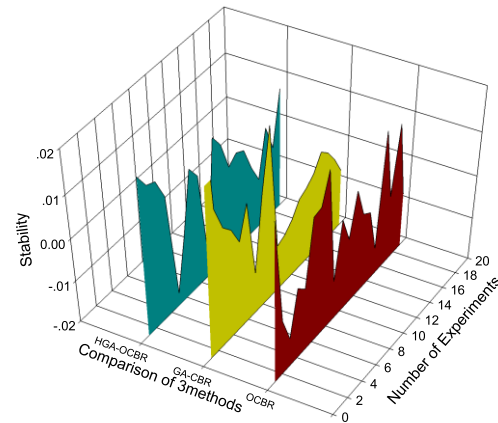


Fig. 11. Accuracy stability of the 3 methods.

The stability of the accuracy is calculated by the slope. The more extreme the peaks and valleys of the line are, the more unstable, and the straighter the line is, the more stable the accuracy. With the increase in the number of traceability experiments, GA-CBR showed only a regular and steady rise and fall, indicating high stability, while OCBR fluctuates greatly and often, indicating general stability. HGA-OCBR has the stability of GA-CBR; there are ups and downs, but they are not extreme.

Figure 12 compares the average traceability times of the three methods. The average traceability time of HGA-OCBR

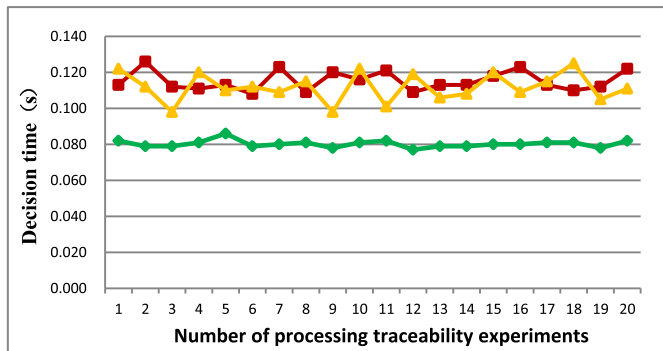


Fig. 12. Average traceability time of the 3 methods.

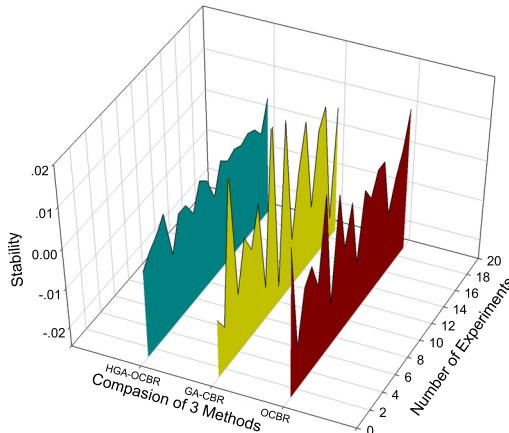


Fig. 13. Traceability speed stability of the 3 methods.

is approximately 0.08 s, which is shorter than the average traceability times of OCBR and GA-CBR.

Figure 13 shows that the traceability speed of GA-CBR fluctuates greatly, indicating that the traceability speed is unstable; OCBR has smaller fluctuations than GA-CBR, but it is also denser and more stable. The traceability speed of HGA-OCBR has little fluctuation and is relatively stable.

In comparison, HGA-OCBR combines the advantages of OCBR and GA-CBR. The case representation through the ontological classification case structure effectively improves the traceability speed and stability, and the GA algorithm is used to improve the case retrieval to improve the traceability accuracy and stability. HGA-OCBR has the best traceability accuracy and traceability speed.

VI. CONCLUSION

By tracing the source code on a product label, the product's origin information, storage, distribution and transportation route, detection time and staff information data can be quickly acquired, but processing link information traceability is rarely involved. Aiming to address the problems of complex product processing and procedures, the high difficulty of collecting and recording information, and error-prone data acquisition, a joint optimization algorithm based on GA and CBR is proposed to establish a process traceability optimization model. This method is meaningful for improving the traceability accuracy of product processing and ensures the integrity of traceability data, which is achieved in three main ways.

- 1) Using an ontology for case representation and an improved GA to optimize case retrieval and reuse, a case reasoning traceability model is constructed to trace all equipment and environmental information in the overall processing in order to trace the complete historical information of the product supply chain.
- 2) OWL-DL syntax is adopted to represent traceability cases and to classify and process information, which facilitates the storage, modification and retrieval of cases and the sharing of case knowledge and domain knowledge, thus simplifying the management and maintenance of the case database.
- 3) This paper not only optimizes the genetic operator by combining the optimal preservation strategy and roulette selection method but also uses the exponential scale transformation method to stretch the fitness function. A joint optimization algorithm is proposed based on GA and CBR, which is used in case retrieval and case reuse.

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